

pySnowClim: fast process-based snow modeling in Python

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Summary

pySnowClim is an open-source Python implementation of the process-based SnowClim model for simulating snow dynamics including accumulation and melt processes. The model achieves a balance between representing important physical processes, such as energy balance calculations, snow density evolution, albedo dynamics, and phase change processes, while simplifying other components to maintain computationally efficiency, delivering consistent snowpack simulations across diverse climatic conditions. Building on these strengths of SnowClim, pySnowClim adopts a layered architecture with vectorized multi-point computation using standardized NetCDF I/O. This makes pySnowClim suitable for point to large scale applications, and ready to couple with climate and hydrological models.

Statement of need

Snow is a critical component of the global water cycle. The accurate simulation of snowpack dynamics is not only essential for water resource management, but also for flood prediction, ecological studies, and climate change impact assessments ([J. T. Anderson, 2023](#); [J. T. Anderson et al., 2025](#); [Dixon et al., 2026](#); [Williams et al., 2026](#)).

The target audience of pySnowClim includes hydrologists, climatologists, ecologists, water resource managers, and students who need reliable snow modeling capabilities for applications such as:

- **Research:** Detailed energy balance studies and process investigations
- **Operations:** Water resource forecasting and management
- **Education:** Teaching snow physics and energy balance concepts
- **Climate Studies:** Long-term snow evolution under changing conditions
- **Adaptation Planning:** Anticipating and planning adaptations to impacts of warming on snow-dependent species, ecosystems, and communities

In addition, the model can be used on different:

- **Spatial Scales:** Point locations to continental domains
- **Temporal Scales:** Sub-daily to daily timesteps, multi-decadal simulations
- **Environments:** Diverse snow climates from maritime to continental

State of the field

Many current snow models are either computationally efficient but only represent physical processes to a very limited extent (e.g. temperature index models), or provide a detailed physical process representation but are too computational burdensome for large-scale high-resolution

38 applications (e.g. most process-based models) (Garen & Marks, 2005; Ikeda et al., 2021; Liston
39 & Elder, 2006; Walter et al., 2005; Wrzesien et al., 2018).

40 SnowClim (Lute et al., 2022) was developed to offer a flexible, efficient, and open source
41 alternative with a good balance between representing physical processes and usability. However,
42 the original MATLAB-based model has no separation of I/O and physics, relies on .mat files, and
43 MATLAB usage is tied to licensing issues, which restricts wider applicability. Thus, pySnowClim
44 being based on Python programming environment, makes the model more accessible to a wider
45 scientific and practical audience when compared to the original MATLAB-based model.

46 Software design

47 In the original MATLAB SnowClim workflow, execution was driven by a wrapper script that
48 prepared data (.mat files) and called the main routine. Parameters were loaded from a file,
49 and the timestep loop computed physics while managing state and outputs. Thus, effectively
50 orchestration and configuration were coupled with the core calculations. In addition, the
51 state of the snowpack used a procedural approach which is more prone to state-update bugs.
52 Additionally, while style works well for a MATLAB-only workflow, it makes it harder to embed
53 SnowClim as a “callable component” inside another model’s time loop without the separation
54 of I/O/configuration from computation.

55 pySnowClim adopts a layered architecture: a core physics engine (timestep calculations) is
56 separated from a runner (I/O + orchestration) and a CLI for batch/operational use. The
57 new implementation also prioritizes vectorized multi-point computation and standardized
58 NetCDF/NumPy I/O, with explicit JSON-based configuration. Research workflows often
59 need the same physics to run in different contexts: notebooks, batch jobs, or pipelines with
60 standardized I/O. Separating the engine from I/O makes easier to validate physics independently
61 of data inputs format.

62 pySnowClim uses an encapsulated state with more abstraction than the procedural approach
63 used in MATLAB, where the model state and outputs are explicit data structures (e.g.,
64 Snowpack state and preallocated SnowModelVariables).

65 Overall, these changes support maintainability, reduce risk of subtle state-update bugs, and
66 make easier the addition of new diagnostics, and variables. The changes allow pySnowClim to
67 be more easily coupled to hydrology/land-surface/ecosystem models as a drop-in component
68 (shared arrays/NetCDF, pipeline execution) without changing core physics.

69 Research impact statement

70 When a model is already published/validated (J. T. Anderson et al., 2025; Boeykens et al., 2025;
71 Dixon et al., 2026; Jans et al., 2025; Williams et al., 2026), researchers need confidence that the
72 new implementation is behaviorally comparable. Also, a well-supported Python package enables
73 easier integration, reproducibility, and further development. pySnowClim provides an extensive
74 documentation including API references, example datasets, and validation against observations
75 from a snow monitoring site. It also includes a comprehensive testing framework, featuring unit
76 tests for individual physics functions and integration tests for complete workflows. In addition,
77 there is a current implementation of coupling between pySnowClim and the Community Water
78 Model (CWatM) (Burek et al., 2020) (<https://github.com/iiasa/CWatM>).

79 The model

80 pySnowClim employs the fundamental principles of mass and energy conservation as its core
81 framework. The model requires meteorological forcing data including temperature, precipitation,
82 shortwave radiation, longwave radiation, wind speed, humidity, and pressure to simulate critical

snow variables such as snow water equivalent (SWE), snow depth, snow density, snow melt, snowpack liquid water content, albedo, and energy fluxes. Figure 1 shows the snow model conceptual diagram.

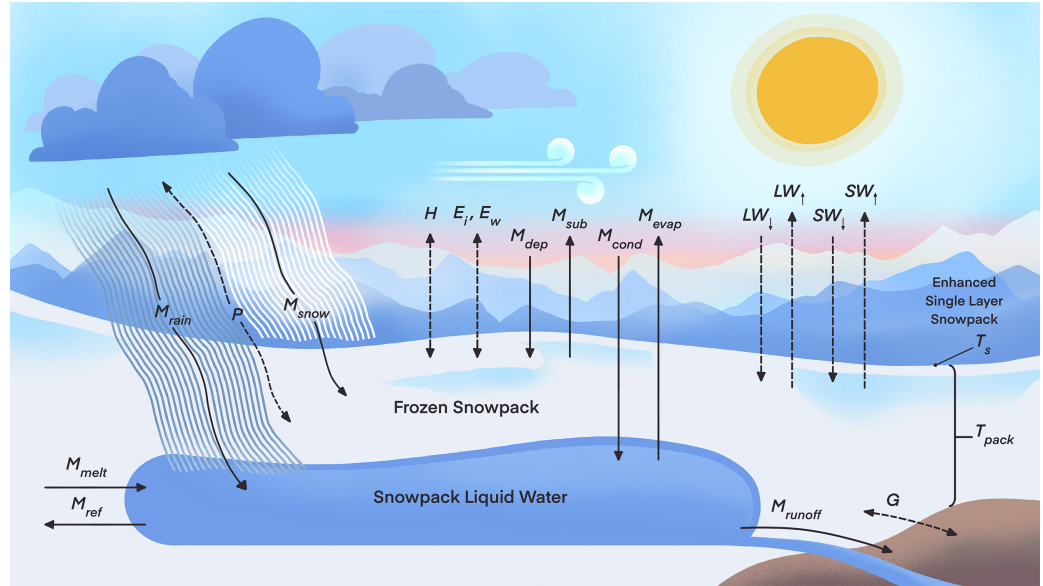


Figure 1: Snow model conceptual diagram. Solid black arrows indicate mass fluxes and dashed black arrows indicate energy fluxes. T_s is snow surface temperature and T_{pack} is the temperature of the snowpack.

pySnowClim calculates the net energy flux to the snow surface accounting for shortwave and longwave radiation, sensible and latent heat fluxes with stability corrections, ground heat flux, and precipitation heat flux Equation 1. The model is built around the surface energy balance equation:

$$Q_{net} = SW_{down} - SW_{up} + LW_{down} - LW_{up} + H + E_i + E_w + G + P \quad (1)$$

Where SW denotes shortwave radiation fluxes, LW denotes longwave radiation fluxes, H is the sensible heat flux, E is the latent heat flux of ice (i) and water (w), G is the ground heat flux, and P is the advected heat flux from precipitation.

The model tracks snow accumulation, sublimation and evaporation processes, snowmelt generation, and liquid water movement through the snowpack with mass conservation Equation 2. Mass balance of the solid (M_s) and liquid (M_l) portions of the snowpack are governed by:

$$M_s = M_{snow} + M_{ref} - M_{melt} + M_{dep} - M_{sub} \quad (2)$$

$$M_l = M_{rain} - M_{ref} + M_{melt} - M_{runoff} + M_{cond} - M_{evap} \quad (3)$$

Where M_{snow} is the mass of new snowfall, M_{ref} is the mass of the snowpack liquid water that has been refrozen, M_{melt} is the mass of snow that has melted, M_{dep} is the mass of deposition, M_{sub} is the mass of sublimation, M_{rain} is the mass of rain added to the snowpack, M_{runoff} is the mass of liquid water that has left the snowpack as runoff, M_{cond} is the mass of condensation, and M_{evap} is the mass of evaporation.

Other key model components include precipitation phase determination using bivariate logistic regression (Jennings et al., 2018), fresh snow density calculation based on temperature (E. A. Anderson, 1976), snow density evolution through compaction processes, liquid water retention and drainage. pySnowClim has three different albedo scheme implementations including aging effects, melting conditions, and seasonal variations with options for different complexity levels (Essery et al., 2013; Hamman et al., 2018; Liang et al., 1994; Tarboton & Luce, 1996). In addition, similarly to other models (E. Anderson, 2006; Burek et al., 2020), a new functionality was added to reduce excessive snow accumulation (i.e. snow towers) using an optional radiation enhancement.

AI usage disclosure

In preparing this work, the authors used ChatGPT (OpenAI) and Claude (Anthropic) to assist with the initial MATLAB-to-Python translation and generation of code documentation. The unit tests of the package were carried out by using Claude. After using these tools, the authors thoroughly reviewed, edited and refined the content, and take full responsibility for the content of this publication. No generative AI tools were used in the development of the writing of this manuscript.

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